In [2]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [3]:
```
sns.set_style("whitegrid")
plt.rcParams['figure.figsize'] = (10, 6)
plt.rcParams['font.size'] = 11
df=pd.read_csv("Cars Datasets 2025.csv", encoding='cp1252')
print(df)
```

```
     Company Names         Cars Names                           Engines  \
0          FERRARI      SF90 STRADALE                                V8
1      ROLLS ROYCE            PHANTOM                               V12
2             Ford               KA+                        1.2L Petrol
3         MERCEDES           GT 63 S                                 V8
4             AUDI        AUDI R8 Gt                                V10
...            ...               ...                                ...
1213        Toyota       Crown Signia                   2.5L Hybrid I4
1214        Toyota   4Runner (6th Gen)  2.4L Turbo I4 (i-FORCE MAX Hybrid)
1215        Toyota      Corolla Cross            2.0L Gas / 2.0L Hybrid
1216        Toyota             C-HR+                 1.8L / 2.0L Hybrid
1217        Toyota     RAV4 (6th Gen)       2.5L Hybrid / Plug-in Hybrid

        CC/Battery Capacity    HorsePower Total Speed  \
0                   3990 cc        963 hp    340 km/h
1                   6749 cc        563 hp    250 km/h
2                  1,200 cc      70-85 hp    165 km/h
3                  3,982 cc        630 hp    250 km/h
4                  5,204 cc        602 hp    320 km/h
...                     ...           ...         ...
1213                2487 cc        240 hp    180 km/h
1214       2393 cc + Battery       326 hp    180 km/h
1215  1987 cc / Hybrid batt  169 – 196 hp    190 km/h
1216  1798 / 1987 cc + batt  140 – 198 hp    180 km/h
1217         2487 cc + batt  219 – 302 hp    200 km/h

     Performance(0 - 100 )KM/H        Cars Prices              Fuel Types  \
0                      2.5 sec        $1,100,000         plug in hyrbrid
1                      5.3 sec          $460,000                  Petrol
2                     10.5 sec    $12,000-$15,000                  Petrol
3                      3.2 sec          $161,000                  Petrol
4                      3.6 sec          $253,290                  Petrol
...                        ...               ...                     ...
1213                   7.6 sec  $43,590 – $48,000  Hybrid (Gas + Electric)
1214                   6.8 sec           $50,000                  Hybrid
1215             8.0 – 9.2 sec  $25,210 – $29,135            Gas / Hybrid
1216            7.9 – 10.5 sec           € 33,000                  Hybrid
1217             6.0 – 8.1 sec  $29,000 – $43,000        Hybrid / Plug-in

     Seats        Torque
0        2        800 Nm
1        5        900 Nm
2        5  100 - 140 Nm
3        4        900 Nm
4        2        560 Nm
...    ...           ...
1213     5        239 Nm
1214     7        630 Nm
1215     5  190 – 210 Nm
1216     5  190 – 205 Nm
1217     5  221 – 400 Nm

[1218 rows x 11 columns]
```

In [4]:
```
cols_to_fix = ['HorsePower', 'Total Speed', 'Cars Prices', 'CC/Battery Capacity']
```

```
for col in cols_to_fix:
    df[col] = df[col].astype(str).str.extract(r'(\d+\.?\d*)').astype(float)
stats = df.describe().loc[['mean', 'std', '50%']]
stats.rename(index={'50%': 'median'}, inplace=True)
print(stats)
```

```
        CC/Battery Capacity  HorsePower  Total Speed  Cars Prices
mean            1413.553009  300.397373   216.467159    68.829088
std             2023.302367  219.343342    53.051077    88.834719
median            82.000000  250.000000   200.000000    40.000000
```

In [5]:
```
df.describe(include='all')
```

Out[5]:

	Company Names	Cars Names	Engines	CC/Battery Capacity	HorsePower	Total Speed	Performance((- 100)KM/I
count	1218	1218	1218	1213.000000	1218.000000	1218.000000	121.
unique	37	1201	356	NaN	NaN	NaN	18(
top	Nissan	KA+	I4	NaN	NaN	NaN	6.5 se
freq	149	2	64	NaN	NaN	NaN	4
mean	NaN	NaN	NaN	1413.553009	300.397373	216.467159	NaI
std	NaN	NaN	NaN	2023.302367	219.343342	53.051077	NaI
min	NaN	NaN	NaN	1.000000	1.000000	80.000000	NaI
25%	NaN	NaN	NaN	2.000000	150.000000	180.000000	NaI
50%	NaN	NaN	NaN	82.000000	250.000000	200.000000	NaI
75%	NaN	NaN	NaN	2000.000000	400.000000	250.000000	NaI
max	NaN	NaN	NaN	16100.000000	2488.000000	500.000000	NaI

In [6]:
```
print(df.columns.tolist())
```

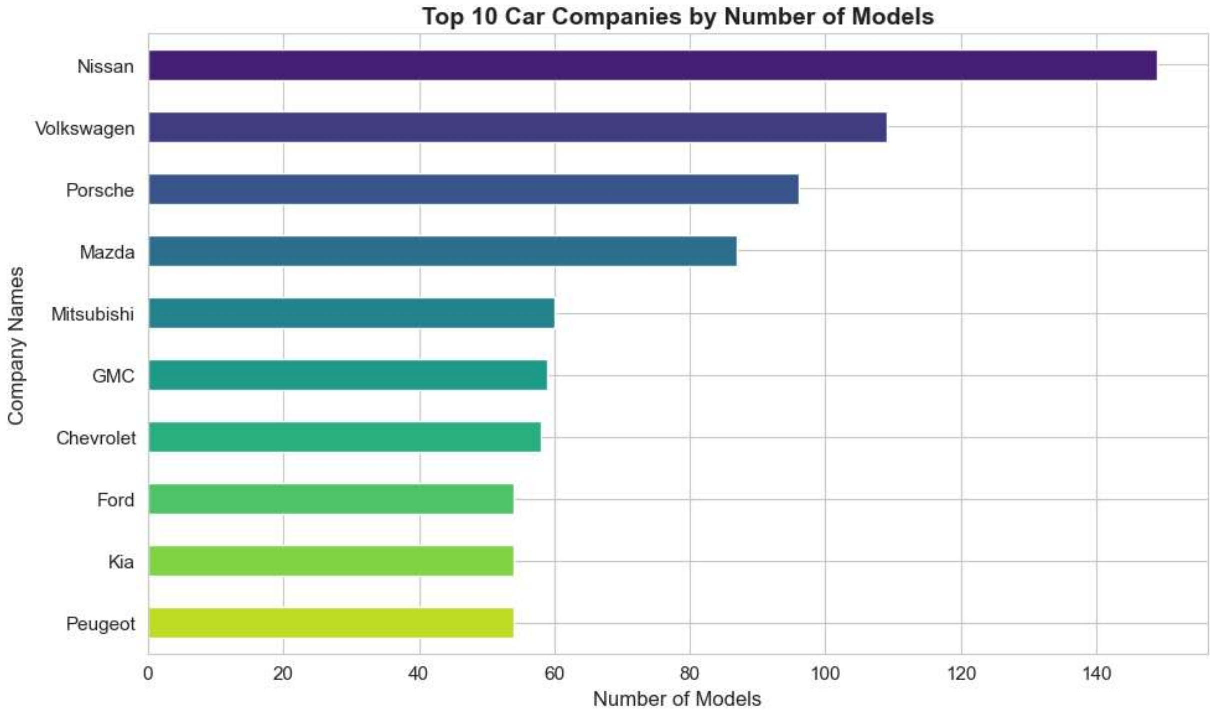
```
['Company Names', 'Cars Names', 'Engines', 'CC/Battery Capacity', 'HorsePower', 'Tot
al Speed', 'Performance(0 - 100 )KM/H', 'Cars Prices', 'Fuel Types', 'Seats', 'Torqu
e']
```

In [7]:
```
print(df.info())
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1218 entries, 0 to 1217
Data columns (total 11 columns):
 #   Column                    Non-Null Count  Dtype  
---  ------                    --------------  -----  
 0   Company Names             1218 non-null   str    
 1   Cars Names                1218 non-null   str    
 2   Engines                   1218 non-null   str    
 3   CC/Battery Capacity       1213 non-null   float64
 4   HorsePower                1218 non-null   float64
 5   Total Speed               1218 non-null   float64
 6   Performance(0 - 100 )KM/H 1212 non-null   str    
 7   Cars Prices               1217 non-null   float64
 8   Fuel Types                1218 non-null   str    
 9   Seats                     1218 non-null   str    
 10  Torque                    1217 non-null   str    
dtypes: float64(4), str(7)
memory usage: 170.1 KB
None
```

1. FEATURE: Company Name (Top Brands)

In [8]:
```
# FEATURE 1: COMPANY Name - Top 10 Brands
plt.figure(figsize=(10, 6))
top_companies = df['Company Names'].value_counts().head(10)
colors = sns.color_palette("viridis", len(top_companies))
top_companies.plot(kind='barh', color=colors)
plt.title('Top 10 Car Companies by Number of Models', fontsize=14, fontweight='bold
plt.xlabel('Number of Models', fontsize=12)
plt.ylabel('Company Names', fontsize=12)
plt.gca().invert_yaxis()
plt.tight_layout()
plt.show()
```



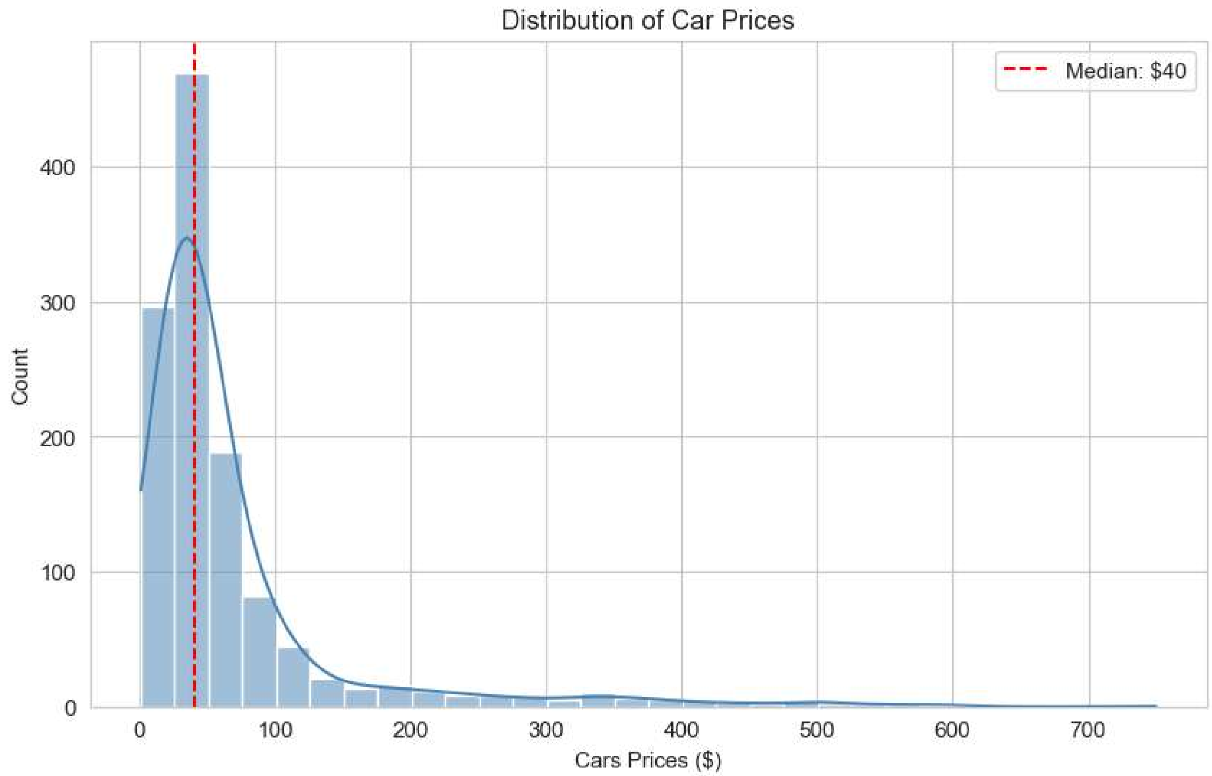
ANSWER:

The bar chart reveals which automotive manufacturers have the most diverse vehicle portfolios. The top companies likely represent major global brands (Toyota, BMW, Mercedes, etc.) that dominate the car market. This helps identify market leaders and understand brand diversity in the dataset. The sorted horizontal format makes it easy to quickly compare company sizes at a glance.

2. FEATURE: Cars Price

In [10]:
```
# FEATURE 3: Cars Price
df['Cars Prices'] = pd.to_numeric(df['Cars Prices'].astype(str).str.replace('[$,]',

plt.figure(figsize=(10, 6))
sns.histplot(df['Cars Prices'].dropna(), bins=30, kde=True, color='steelblue')
plt.axvline(df['Cars Prices'].median(), color='red', linestyle='--',
            label=f'Median: ${df["Cars Prices"].median():,.0f}')
plt.title('Distribution of Car Prices')
plt.xlabel('Cars Prices ($)')
plt.legend()
plt.show()
```



ANSWER:

The histogram with KDE shows how car prices are distributed across the market. Notice the right-skewed distribution—most cars cluster at lower price points with a long tail of luxury vehicles. The red dashed line marks the median price, giving you a central reference point. This visual helps identify price tiers: budget, mid-range, and luxury segments, and shows if there are distinct price clusters that might represent different vehicle categories.

3. FEATURE: Fuel Type Comparison

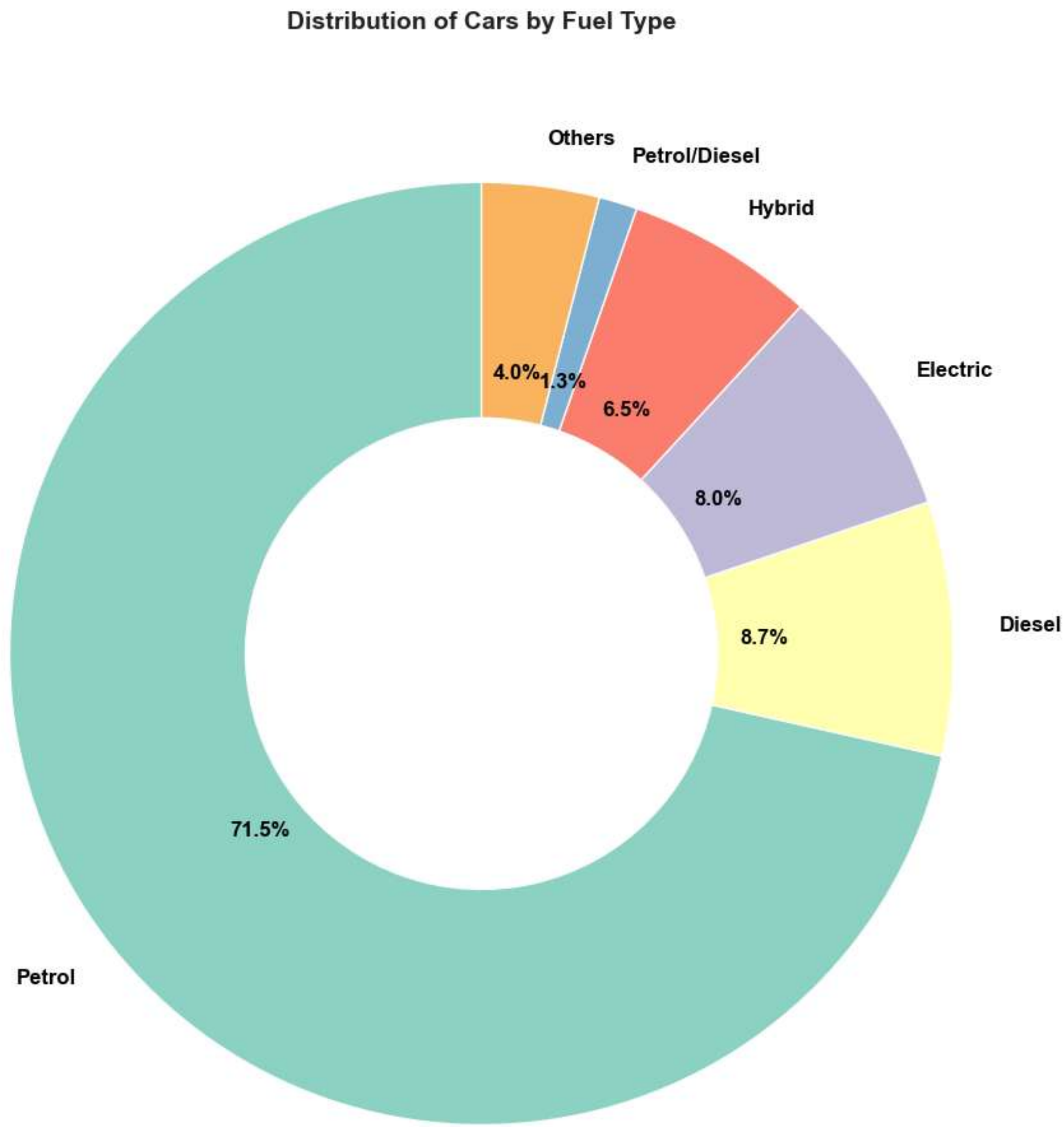
In [11]:

```
# FEATURE 3: FUEL TYPE - Proportions
plt.figure(figsize=(10, 10))
fuel_counts = df['Fuel Types'].value_counts()
top_5 = fuel_counts.head(5)
others = fuel_counts.iloc[5:].sum() if len(fuel_counts) > 5 else 0
if others > 0:
    top_5['Others'] = others

colors = sns.color_palette("Set3", len(top_5))
plt.pie(top_5,
        labels=top_5.index,
        autopct='%1.1f%%',
```

```
        colors=colors,
        startangle=90,
        wedgeprops=dict(width=0.5, edgecolor='white'),
        textprops={'fontsize': 12, 'color': 'black', 'weight': 'bold'})  # Control

plt.title('Distribution of Cars by Fuel Type', fontsize=14, fontweight='bold', pad=
centre_circle = plt.Circle((0,0), 0.3, fc='white')
plt.gca().add_artist(centre_circle)
plt.tight_layout()
plt.show()
```



ANSWER:

```
    Reveals market share by fuel technology. The largest slice
identifies the dominant fuel type (traditional combustion vs.
electric/hybrid), showing current automotive industry trends
```

4. FEATURE: HorsePower vs Total Speed

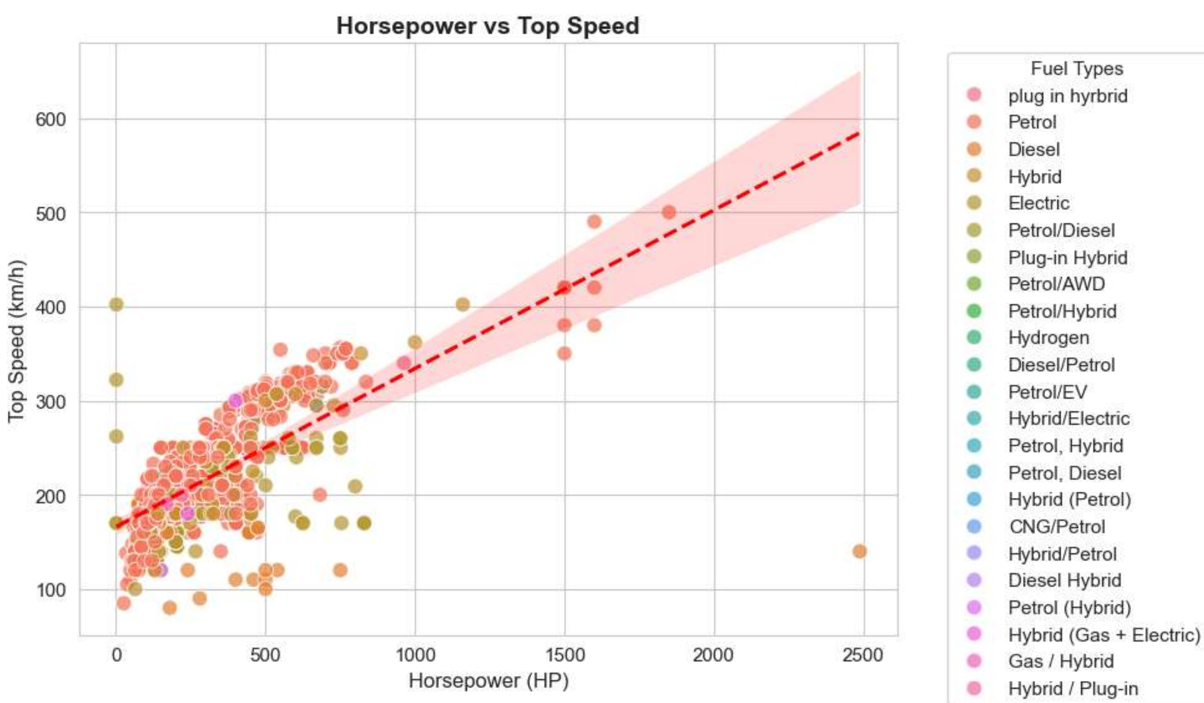
ANSWER:

```
    Positive correlation between horsepower and top speed.
Colored dots show which fuel types deliver best speed-to-power
ratio. Red line marks the trend.
```

In [12]:
```
# Feature 4: HorsePower vs Total Speed
plt.figure(figsize=(10, 6))

sns.scatterplot(data=df, x='HorsePower', y='Total Speed', hue='Fuel Types', alpha=0
sns.regplot(data=df, x='HorsePower', y='Total Speed', scatter=False, color='red', l

plt.title('Horsepower vs Top Speed', fontsize=14, fontweight='bold')
plt.xlabel('Horsepower (HP)', fontsize=12)
plt.ylabel('Top Speed (km/h)', fontsize=12)
plt.legend(title='Fuel Types', bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
plt.show()
```

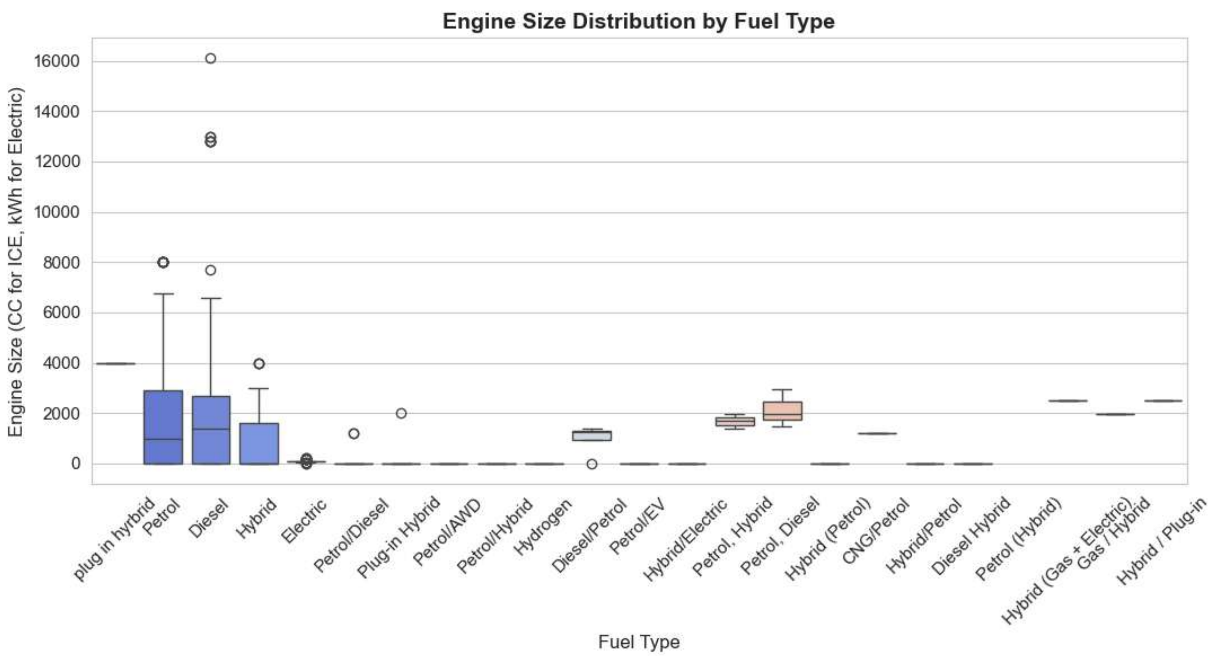


5. FEATURE:CC/Battery Capacity

In [13]:
```
# Feature 5: Engine Size Distribution by Fuel Type
df['CC/Battery Capacity'] = pd.to_numeric(df['CC/Battery Capacity'], errors='coerce
```

```python
plt.figure(figsize=(11, 6))
sns.boxplot(data=df, x='Fuel Types', y='CC/Battery Capacity',
         hue='Fuel Types', palette='coolwarm', legend=False)

plt.title('Engine Size Distribution by Fuel Type', fontsize=14, fontweight='bold')
plt.xlabel('Fuel Type', fontsize=12)
plt.ylabel('Engine Size (CC for ICE, kWh for Electric)', fontsize=12)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



ANSWER:

    Box shows middle 50% of values, line is median. Compares CC vs kWh across fuel types.